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Dual-Loop Voltage–Current Control of a Fractional-Order Buck-Boost Converter Using a Fractional-Order PI^λ Controller

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Abstract: Based on the fact that the inductor and capacitor are of a non-integer order by nature, to provide a more accurate theoretical basis for the optimal control of the converter, the fractional-order model of the Buck-Boost converter in the continuous mode of current is established according to the fractional-order calculus theory. The fractional-order PI^λ control system of the fractional-order Buck-Boost converter is designed to compare the performance of the integer-order PI controller with the fractional-order controller. Secondly, the sparrow search algorithm is applied to the optimal design of the fractional-order PI^λ control system of the fractional-order Buck-Boost converter to improve the system's phase margin, stability, and robustness. Finally, the simulation is verified on the Matlab/Simulink simulation platform and compared with the integer-order PI controller.

Keywords: Buck-Boost converters; fractional-order; PI^λ controller; SSA



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1. Introduction

Every system in engineering exists as a fractional-order system. To avoid complexity and consider ease of use, the system is approximated in an integer-order form, which, inevitably, involves a certain amount of error. In power electronic converter circuits, capacitors and inductors are fundamental electronic devices. Their actual external characteristic equations have fractional-order properties; thus, inductors and capacitors are both fractional-order in nature [1–3]. A DC converter is a power electronic device that converts DC energy into the DC energy of a controlled voltage or current required by the load. It works by controlling the on/off of the DC converter circuit through a fully controllable device, i.e., a high-frequency switch, to regulate the input voltage so that the desired DC voltage can be output. DC converters are widely used in various fields, including industry, vehicles, and renewable energy, due to the reliability of their output [4–6]. Therefore, it is important to establish and analyze an accurate mathematical model of the DC converter. Based on the fact that inductors and capacitors are fractional-order in nature, describing and analyzing power electronic converters with an integer-order model is not accurate enough and may even be wrong. Therefore, it is necessary to establish a fractional-order model of the converter, which can improve the accuracy of modeling and more accurately describe its actual dynamic behavior and provide a more accurate theoretical basis for the optimal control of the converter.

With the rapid development of information, science, and technology, fractional order calculus theory has been more widely used in automatic control. The fractional-order controller has more advantages than the traditional integer-order controller, and the complete fractional-order system has better dynamic performance and robustness [7]. The fractional-order $PI^\lambda D^\mu$ control is now widely used in machinery [8,9] and automobiles [10,11]. Reference [12] built the fractional-order circuit model and mathematical model of the Boost converter in a Matlab/Simulink environment and designed a fractional-order PI^λ controller

to control the fractional-order system to form a fully fractional-order system. Reference [13] established a fractional-order model of the Boost converter in the pseudo-continuous mode of inductor current and designed a fractional-order nonlinear controller to improve the dynamic and steady-state performance of the fractional-order system. Reference [14] proposed a PI^λ controller based on the Gray Wolf algorithm as a method for power factor correction of the internal current controller and the external voltage controller and regulation of the load voltage and compared it with the conventional PI controller. The simulation results showed the better performance of the PI^λ controller based on the Gray Wolf algorithm. Reference [15] investigated the effect of the fractional-order PI^λ controller on the control performance of the Buck converter. It analyzed the impact of the PI^λ controller on the steady-state performance of the system and the stability range of the proportional and integral coefficients when λ takes different values under the premise of ensuring the stability of the system.

The Sparrow Search Algorithm (SSA) is mainly inspired by the foraging and anti-predation behavior of sparrows. The algorithm is relatively novel, with the advantages of a strong optimization ability and a fast convergence speed [16]. In this paper, the SSA algorithm is used to optimize the controller parameters and optimize the control performance of the control system. This paper is organized as follows. Section 2 uses the fractional-order Caputo calculus definition to build the fractional-order state averaging model of the Buck-Boost converter and find the transfer function of the converter. In Section 3, the fractional-order PI^λ controller is designed, the fractional-order PI^λ control system model of the fractional-order Buck-Boost converter is built in the Matlab/Simulink simulation platform, and the performance of the fractional-order PI^λ control system and the integer-order PI control system is compared. In Section 4, SSA is used to perform a parameter search for the fractional-order PI^λ controller, and SSA is applied to the optimal design of the fractional-order Buck-Boost PI^λ control system to improve the system performance. The effectiveness of the method is verified by simulation. Finally, conclusions are given in Section 5.

2. State Averaging Model for Fractional-Order Buck-Boost Converters

The circuit topology of the fractional-order Buck-Boost converter is shown in Figure 1. In the figure, V_{in} is the power supply voltage, S_T and S_D are the switching tubes, R is the circuit resistance, and i_L and v_C are the inductor current and capacitor voltage, respectively. α and β are the orders of the inductor and capacitor, respectively.

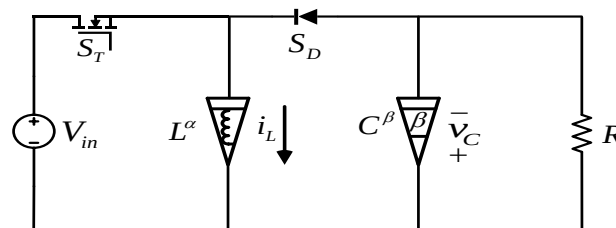


Figure 1. Circuit diagram of fractional-order Buck-Boost converter.

According to Reference [17], the mathematical expressions for the fractional-order inductor current i_L and capacitor voltage v_C are obtained as:

$$\begin{cases} v_L(t) = L \frac{d^\alpha i_L}{dt^\alpha} \\ i_C(t) = C \frac{d^\beta v_C}{dt^\beta} \end{cases} \quad (1)$$

where α and β are the fractional order inductance and capacitance orders, respectively, and $0 < \alpha < 1, 0 < \beta < 1$.

The fractional-order CCM Buck-Boost converter has two operating modes within one switching cycle T_s .

Mode 1 ($0 < t \leq DT_s$): Switching tube S_T is on, and the diode S_D is off.

$$\begin{cases} \frac{d^\alpha i_L}{dt^\alpha} = \frac{V_{in}}{L} \\ \frac{d^\beta v_C}{dt^\beta} = -\frac{v_0}{RC} \end{cases} \quad (2)$$

Mode 2 ($DT_s < t \leq T_s$): Switching tube S_T is off, and the diode S_D is on.

$$\begin{cases} \frac{d^\alpha i_L}{dt^\alpha} = -\frac{v_0}{L} \\ \frac{d^\beta v_C}{dt^\beta} = \frac{i_L}{C} - \frac{v_0}{RC} \end{cases} \quad (3)$$

where D is the duty cycle.

In this paper, the fractional-order Caputo calculus is used to define the fractional-order state averaging model of the Buck-Boost converter. The uniform fractional order calculus operator ${}_t D^\alpha$ is defined as [18]:

$${}_t D^\alpha f(t) = \begin{cases} \frac{d^\alpha}{dt^\alpha} f(t) & \alpha > 0 \\ f(t) & \alpha = 0 \\ \int_{t_0}^t f(\tau) d\tau^{-\alpha} & \alpha < 0 \end{cases} \quad (4)$$

where α is the order of the fractional order ($\alpha \in \mathbb{R}$). t and t_0 are the upper and lower bounds of the integration, respectively. When $\alpha > 0$ and α is a real number, ${}_t D^\alpha$ represents fractional-order differentiation; when $\alpha < 0$ and α is a real number, ${}_t D^\alpha$ represents a fractional-order integral.

The Caputo differential is defined as [19]:

$${}_t D^\alpha f(t) = \frac{1}{\Gamma(m-\alpha)} \int_{t_0}^t \frac{f^{(m)}(\tau)}{(t-\tau)^{1+\alpha-m}} d\tau \quad (5)$$

The Caputo integral is defined as:

$${}_t D^{-\alpha} f(t) = \frac{1}{\Gamma(\alpha)} \int_{t_0}^t \frac{f(\tau)}{(t-\tau)^{-\alpha+1}} d\tau \quad (6)$$

where $m \in \mathbb{Z}^+$ and $m-1 < \alpha < m$.

Using the state-space averaging method [20], the state variables are averaged over a switching period, as follows.

$$\langle x(t) \rangle_T = \frac{1}{T} \int_t^{T+t} x(\tau) d\tau \quad (7)$$

where $x(t)$ represents the state variable in the equation of state. When averaging the fractional order differentiation of state variables, there is the following equation:

$$\frac{d^\alpha \langle x(t) \rangle_T}{dt^\alpha} = \frac{d^\alpha \left(\frac{1}{T} \int_t^{T+t} x(\tau) d\tau \right)}{dt^\alpha} = \frac{1}{T} \int_t^{T+t} \frac{d^\alpha x(\tau)}{dt^\alpha} d\tau = \left\langle \frac{d^\alpha x(t)}{dt^\alpha} \right\rangle \quad (8)$$

From Equation (8), it can be seen that the fractional differential of the state mean variable is equivalent to the average of the fractional order differential of the state variable.

Therefore, the fractional order differential equation under two different operating modes can be converted into an equation of the average variable for representation.

The fractional-order state averaging model of the Buck-Boost converter under CCM is derived by averaging Equations (2) and (3) over one switching period T , based on the fractional-order calculus property and the state-averaging modeling method. It is shown as follows:

$$\begin{cases} \frac{d^\alpha \langle i_L \rangle}{dt^\alpha} = \frac{d \langle V_{in} \rangle}{L} - \frac{\langle v_0 \rangle (1-d)}{L} \\ \frac{d^\alpha \langle v_0 \rangle}{dt^\alpha} = \frac{(1-d) \langle i_L \rangle}{C} - \frac{\langle v_0 \rangle}{RC} \end{cases} \quad (9)$$

where $\langle i_L \rangle$, $\langle V_{in} \rangle$, and $\langle v_0 \rangle$ denote the average values of the corresponding variables in a switching cycle. d is the duty cycle of the switching tube. $\langle i_L \rangle$, $\langle V_{in} \rangle$, $\langle v_0 \rangle$, and d in Equation (9) can be decomposed into dc components and small signal perturbation terms in the following way:

$$\begin{cases} \langle i_L \rangle = I_L + \hat{i}_L \\ \langle v_{in} \rangle = V_{in} + \hat{v}_{in} \\ \langle v_0 \rangle = v_0 + \hat{v}_0 \\ d = D + \hat{d} \end{cases} \quad (10)$$

where I_L , V_{in} , v_0 , and D represent the steady-state direct flows of the corresponding variables. $\hat{i}_L$, $\hat{v}_{in}$, $\hat{v}_0$, and $\hat{d}$ denote the small signal perturbation sections of the corresponding variables. By substituting Equation (10) into Equation (9) and neglecting the second-order terms, the averaging-model form is shown below.

$$\begin{cases} \frac{d^\alpha (I_L + \hat{i}_L)}{dt^\alpha} = \frac{(D + \hat{d})(V_{in} + \hat{v}_{in})}{L} - \frac{(v_0 + \hat{v}_0)(1 - D - \hat{d})}{L} \\ \frac{d^\beta (v_0 + \hat{v}_0)}{dt^\beta} = \frac{(1 - D - \hat{d})(I_L + \hat{i}_L)}{C} - \frac{(v_0 + \hat{v}_0)}{RC} \end{cases} \quad (11)$$

According to the definition of fractional-order calculus, the separation of the direct and AC quantities of the fractional-order CCM Buck-Boost converter in Equation (11) is obtained as follows:

$$\begin{cases} L \left[\frac{d^\alpha I_L}{dt^\alpha} + \frac{d^\alpha \hat{i}_L}{dt^\alpha} \right] = DV_{in} - v_0(1 - D) + D\hat{v}_{in} + (V_{in} + v_0)\hat{d} + \hat{v}_0(1 - D) \\ RC \left[\frac{d^\beta v_0}{dt^\beta} + \frac{d^\beta \hat{v}_0}{dt^\beta} \right] = (1 - D)RI_L - v_0 + (1 - D)\hat{i}_L - \hat{v}_0 - RI_L\hat{d} \end{cases} \quad (12)$$

Removing the direct flow in Equation (12), the expression for the AC volume of the converter is obtained as:

$$\begin{cases} L \frac{d^\alpha \hat{i}_L}{dt^\alpha} = D\hat{v}_{in} + (V_{in} + v_0)\hat{d} + \hat{v}_0(1 - D) \\ RC \frac{d^\beta \hat{v}_0}{dt^\beta} = (1 - D)\hat{i}_L - \hat{v}_0 - RI_L\hat{d} \end{cases} \quad (13)$$

Using the Caputo definition of fractional-order calculus to obtain the Laplace transform of Equation (13) is shown in Equation (14) [21].

$$\begin{cases} Ls^\alpha \hat{i}_L(s) = D\hat{v}_{in}(s) + (V_{in} + v_0)\hat{d}(s) + \hat{v}_0(s)(1 - D) \\ Cs^\beta \hat{v}_0(s) = \frac{(1-D)}{R} \hat{i}_L(s) - \frac{1}{R} \hat{v}_0(s) - I_L \hat{d}(s) \end{cases} \quad (14)$$

The expression of the transfer function of the output voltage of the converter to the inductor current can be obtained as:

$$G_{vi}(s) = \frac{\hat{v}_0}{\hat{i}_L} = \frac{R(1 - D)(V_{in} + v_0) - I_L L R s^\alpha}{(C R s^\beta + 1)(V_{in} + v_0) + I_L(1 - D)R} \quad (15)$$

The expression for the transfer function of the inductor current to duty cycle of the converter is:

$$G_{id}(s) = \frac{\hat{i}_L}{\hat{d}} = \frac{(CRs^\beta + 1)(V_{in} + v_0) + R(1 - D)I_L}{CRLs^{\alpha+\beta} + Ls^\alpha + R(1 - D)^2} \quad (16)$$

From Equations (15) and (16), we can see that the transfer function of the converter is related to the order of the energy storage element. When the order of the energy storage element $\alpha = 1$ and $\beta = 1$, the transfer function of the fractional-order converter is the same as that of the integer-order converter, which shows that the integer-order converter is only a special case of the fractional-order converter, and the fractional-order calculus theory can describe the characteristics of the DC–DC converter more accurately.

3. Design of Fractional-Order PI^λ Controller

3.1. Introduction to Fractional-Order PI^λ Controller

The fractional-order PI^λ controller is a further improvement on the traditional PI controller. The fractional-order PI^λ controller has one more fractional-order parameter than the conventional integer-order PI controller, i.e., the fractional-order parameter λ of integration. This parameter change allows the controller to take any value on a straight line, equivalent to changing the parameter value of a conventional PI controller of integration order 1 from a fixed point to all points on a straight line [22]. In this section, the fractional-order PI^λ controller is designed.

The basic framework of the fractional-order PI^λ control system is shown in Figure 2.

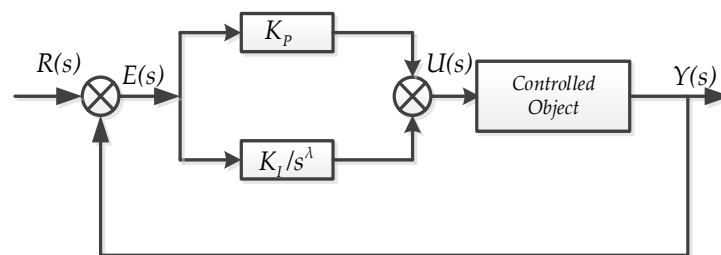


Figure 2. Block diagram of fractional order PI^λ control system.

The transfer function of the fractional-order PI^λ controller can be obtained from Figure 2 as:

$$G(s) = K_p + K_I \frac{1}{s^\lambda} \quad (17)$$

The output expression of the fractional-order PI^λ controller in the time domain is:

$$u(t) = K_p e(t) + K_I D^{-\lambda} e(t) \quad (18)$$

The relationship between the IOPID and FOPID controllers are presented in Figure 3 [23]. Figure 3a shows the values of each parameter of the integer-order controller, and Figure 3b shows the distribution of the positions of the fractional-order controller parameter points that can be placed on the horizontal axis λ .

As can be seen from Figure 3, each parameter of the conventional PID controller can only be selected at four fixed points. When λ and μ are both equal to 0, it is a P controller; when λ is 0 and μ is 1, it is a PD controller; when λ is 1 and μ is 0, it is a PI controller; when λ and μ are both equal to 1, it is an integer-order PID controller. The value of the λ parameter of the fractional-order PI^λ controller can choose any point on the horizontal axis straight line, as shown in Figure 3b, and the integration order λ can be any position on the positive half-axis of λ if the controlled system can be kept in the stable range, thus the form of fractional-order controller has better flexibility.

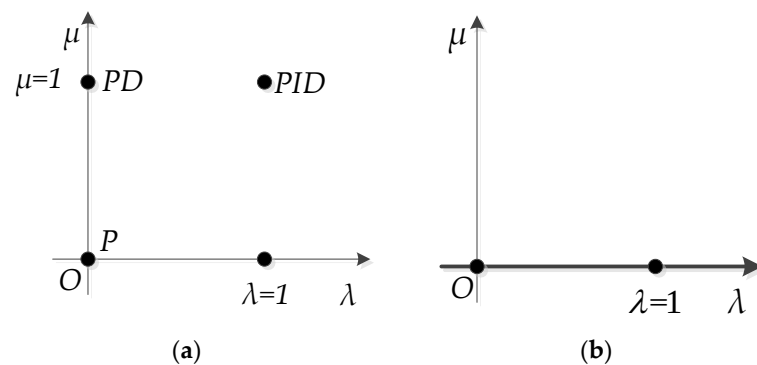


Figure 3. Controller parameter distribution diagram. (a) Integer order. (b) Fractional Order.

3.2. Design of Fractional-Order PI^λ Controller

The transfer function of the converter and the fractional PI^λ controller are combined to establish a double closed-loop system with a fractional-order Buck-Boost converter current inner loop and voltage outer loop, as shown in Figure 4. $G_{vc}(s)$ is the voltage loop fractional-order PI^λ controller, $G_{ic}(s)$ is the current loop fractional-order PI^λ controller, $G_{id}(s)$ is the fractional-order transfer function of the inductor current to duty cycle, and $G_{vi}(s)$ is the fractional-order transfer function of the output voltage to inductor current. The expressions are given in Equations (15) and (16).

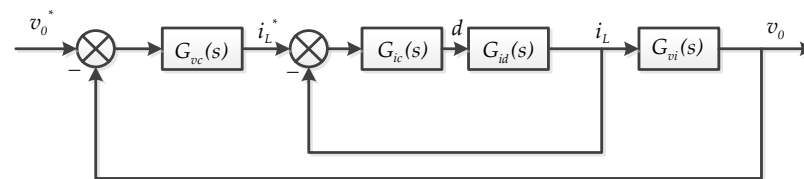


Figure 4. PI^λ control block diagram of fractional order buck boost converter.

The current-loop open-loop transfer function $G_{ii}(s)$ and the current-loop closed-loop transfer function $G_{io}(s)$ of the fractional-order Buck-Boost converter control system under the control of the current inner loop and the voltage outer loop are:

$$\begin{cases} G_{ii}(s) = G_{ic}(s)G_{id}(s) = (K_{PI} + \frac{K_{II}}{s^{\lambda_i}}) \frac{(CRs^\beta + 1)(V_{in} + v_0) + R(1-D)I_L}{CRLs^{\alpha+\beta} + Ls^\alpha + R(1-D)^2} \\ G_{io}(s) = \frac{G_{ii}(s)}{1 + G_{ii}(s)} = \frac{(K_{PI} + \frac{K_{II}}{s^{\lambda_i}}) \frac{(CRs^\beta + 1)(V_{in} + v_0) + R(1-D)I_L}{CRLs^{\alpha+\beta} + Ls^\alpha + R(1-D)^2}}{1 + (K_{PI} + \frac{K_{II}}{s^{\lambda_i}}) \frac{(CRs^\beta + 1)(V_{in} + v_0) + R(1-D)I_L}{CRLs^{\alpha+\beta} + Ls^\alpha + R(1-D)^2}} \end{cases} \quad (19)$$

Combining Equations (15), (17), and (19), the system open-loop transfer function $G_{vi}(s)$ and the closed-loop transfer function $G_{vo}(s)$ can be obtained as:

$$\begin{cases} G_{vi}(s) = G_{vc}(s)G_{io}(s)G_{vi}(s) = (K_{PV} + \frac{K_{IV}}{s^{\lambda_2}}) \frac{R(1-D)(V_{in} + v_0) - I_L L R s^\alpha}{(CRs^\beta + 1)(V_{in} + v_0) + I_L(1-D)R} \\ G_{vo}(s) = \frac{G_{vc}(s)G_{io}(s)G_{vi}(s)}{1 + G_{vc}(s)G_{io}(s)G_{vi}(s)} = \frac{(K_{PV} + \frac{K_{IV}}{s^{\lambda_2}}) \frac{R(1-D)(V_{in} + v_0) - I_L L R s^\alpha}{(CRs^\beta + 1)(V_{in} + v_0) + I_L(1-D)R}}{1 + [(K_{PV} + \frac{K_{IV}}{s^{\lambda_2}}) \frac{R(1-D)(V_{in} + v_0) - I_L L R s^\alpha}{(CRs^\beta + 1)(V_{in} + v_0) + I_L(1-D)R}]} \end{cases} \quad (20)$$

In Equation (20), K_{PI} and K_{II} are the proportionality and integration coefficients of the current loop controller, respectively, and K_{PV} and K_{IV} are the proportionality and integration coefficients of the voltage loop controller, respectively. λ_i and λ_v are the integration orders of the current and voltage loop controllers, respectively. The α and β fractions are the orders of the inductive and capacitive components, respectively, both of which are 0.9. I_L and v_0 are the direct current flow and output voltage, respectively. Power supply voltage $V_{in} = 25$ V, load resistance $R = 80 \Omega$, switch duty cycle $D = 0.6$, switching frequency $f_s = 50$ KHz, fractional inductance $L = 5$ mH, and fractional capacitor $C = 0.1$ mF/(s) $^{1-\beta}$.

Using the Ziegler–Nichols rule to adjust the parameters of the fractional-order PI $^\lambda$ controller, the parameters of the current loop fractional-order PI $^\lambda$ controller and integer-order PI controller are obtained as shown in Table 1.

Table 1. Fractional order PI $^\lambda$ and integer order PI controller parameters of current loop.

Controller	K_{IP}	K_{II}	λ_i
Integer order	0.04	9.26	1
Fractional order	0.04	9.26	0.9

Its voltage loop fractional-order PI $^\lambda$ controller and integer-order PI controller parameters are shown in Table 2.

Table 2. Fractional order PI $^\lambda$ and integer order PI controller parameters of voltage loop.

Controller	K_{IP}	K_{II}	λ_i
Integer order	0.04	9.26	1
Fractional order	0.04	9.26	0.9

Based on the open-loop transfer functions obtained from Equations (16) and (17), the open-loop frequency characteristics of the fractional-order Buck-Boost converter system are shown in Figure 5 when comparing the control of integer-order PI and fractional-order PI $^\lambda$ using a modified Oustaloup filter for the fractional-order calculus operator.

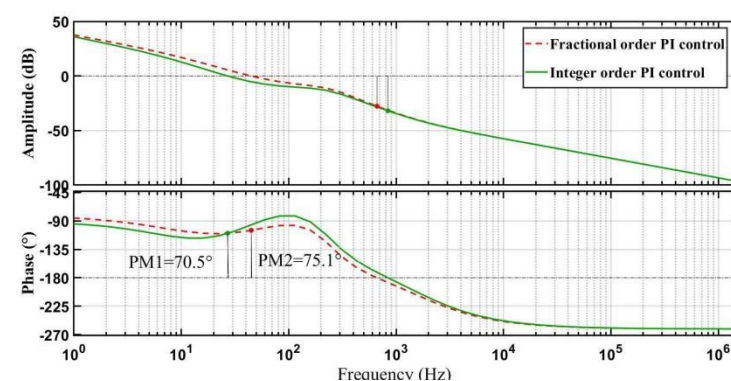


Figure 5. Open loop frequency characteristic diagram of two controllers.

As can be seen from Figure 5, the red dashed line and the green solid line are the logarithmic curves of the frequency characteristics of the fractional-order PI $^\lambda$ control system and the integer-order PI control system, respectively. When an integer-order PI controller is used, the phase margin of the system is 70.5° and its shear frequency is 30.7 Hz; when a fractional-order PI $^\lambda$ control is used, the phase margin of the system is increased to 75.1° and the shear frequency of the system is increased to 42.7 Hz. There is a corresponding increase in the phase margin and shear frequency when using the fractional-order controller. Therefore, the use of a fractional-order controller makes the tracking performance of the system stronger and the transition process shorter, which improves the stability of the system.

3.3. Control System Simulation and Analysis

Based on the Oustaloup frequency domain filtering method in the Matlab/Simulink simulation platform to build a fractional-order PI^λ control system model of a fractional-order Buck-Boost converter [24,25], the output current and output voltage of the converter are studied by changing the load resistance of the converter. The performance of the fractional-order PI^λ control system and the integer-order PI control system are compared. The simulation circuit diagram is shown in Figure 6. In Figure 6, the equivalent circuit method is used to construct fractional-order inductors and fractional-order capacitors of a specific order for investigation. When the fitting order is 9th, the equivalent circuit of the fractional-order inductor and fractional-order capacitor is shown in Figure 7.

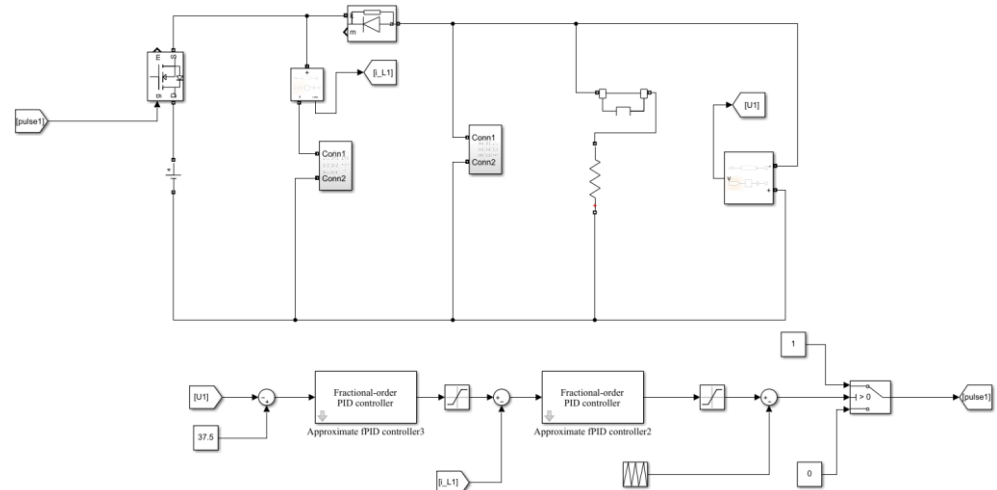


Figure 6. Simulation circuit diagram.

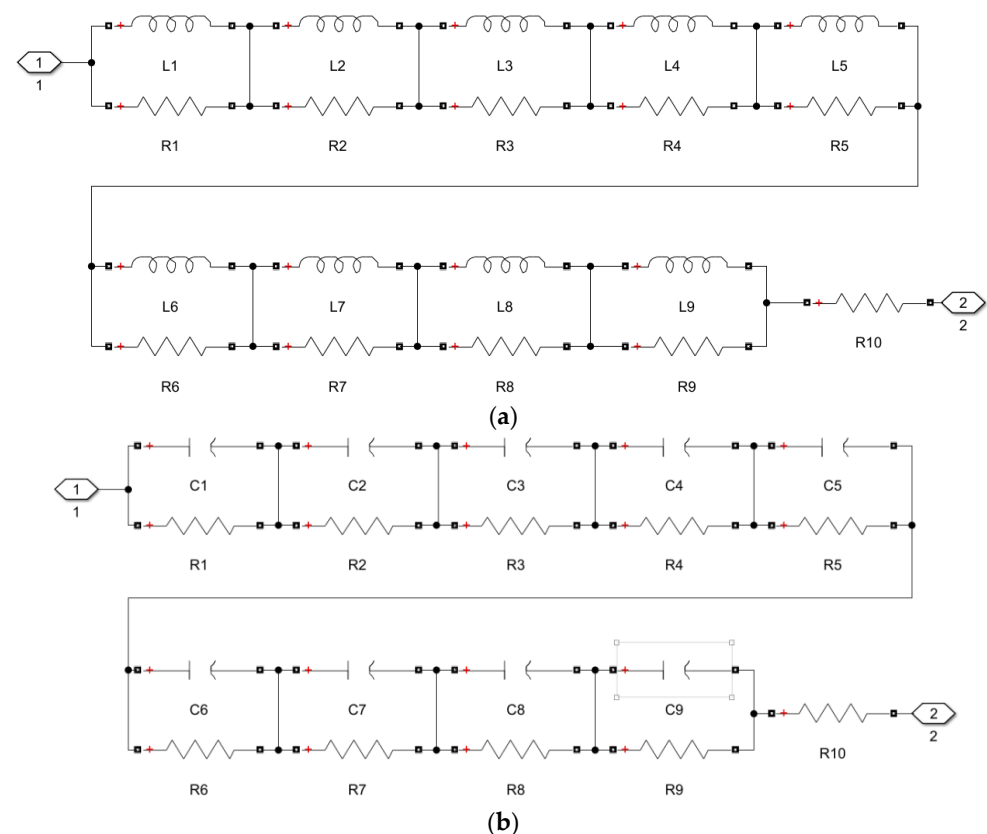


Figure 7. Equivalent circuit of fractional element. (a) Fractional-order inductor. (b) Fractional-order capacitor.

The simulation parameters of the system are shown in Table 1. To verify the stability and anti-interference performance of the system, the resistive load is changed at the simulation time of 0.25 s, and the load resistance is abruptly changed from 160 Ω to 80 Ω . The simulation results are shown in Figure 8.

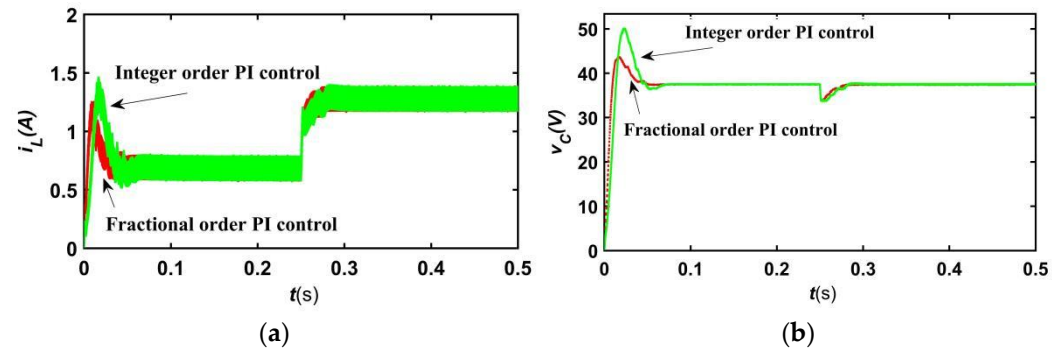


Figure 8. Comparison diagram of system inductance current and capacitance voltage. (a) Inductor current. (b) Capacitor voltage.

From Figure 8, it can be seen that the fractional-order PI $^\lambda$ control system is better than the integer-order PI control system in terms of rise time, peak time, and anti-interference performance, and the local simulation enlargement is shown in Figures 9 and 10.

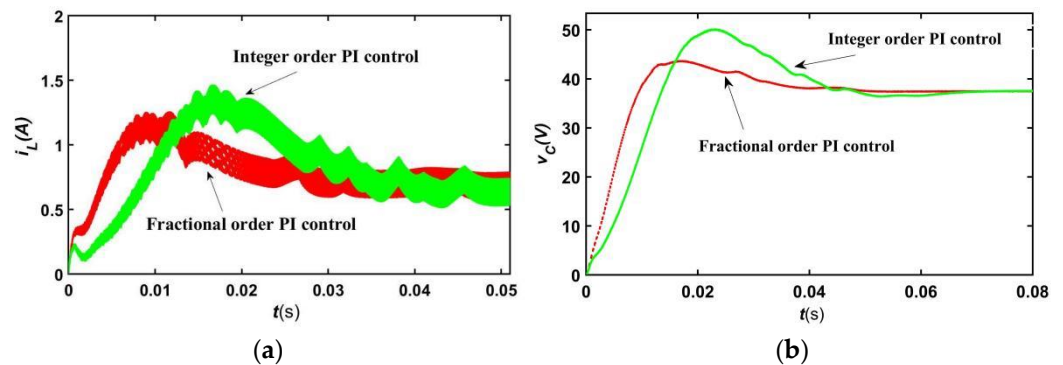


Figure 9. Enlarged view of inductor current and capacitor voltage in initial rising stage. (a) Inductor current. (b) Capacitor voltage.

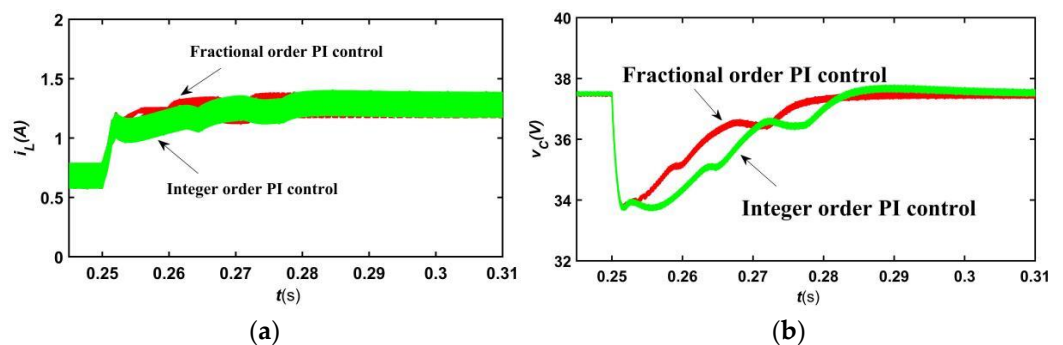


Figure 10. Enlarged drawing of inductor current and capacitor voltage under sudden load change. (a) Inductor current. (b) Capacitor voltage.

According to the simulation results of Figures 8–10, the process of the initial rise phase and sudden load change can be obtained, and the comparison of each performance index of the two systems is shown in Tables 3 and 4.

Table 3. Comparison of system response index of inductive current under two controllers.

Status	Controller	Rise Time (s)	Reduction Ratio of Adjustment Time (%)	Maximum Fluctuation Amount (A)	Maximum Volatility Reduction Ratio (%)
Initial state	PI	0.0041	—	0.663	—
	PI ^λ	0.0087	21.3%	0.461	30.46%
Sudden load change	PI	0.0291	—	0	—
	PI ^λ	0.0234	19.6%	0	0

Table 4. Comparison of system response index of capacitor voltage under two controllers.

Status	Controller	Rise Time (s)	Reduction Ratio of Adjustment Time (%)	Maximum Fluctuation Amount (A)	Maximum Volatility Reduction Ratio (%)
Initial state	PI	0.0136	—	12.62	—
	PI ^λ	0.0095	25.4%	6.13	51.43%
Sudden load change	PI	0.03	—	3.83	—
	PI ^λ	0.0232	22.67%	3.8	0.78%

Figures 8–10 show that the fractional-order PI^λ controller shortens the rise time and regulation time of the system, and the regulation time of the initial state of the capacitor voltage is reduced by 25.4%, and the maximum fluctuation is reduced by 51.42%. After the disturbance, the regulation time is shortened by 22.67%, the maximum fluctuation is reduced by 0.78%, and the system can quickly reach a new steady state. Thus, the fractional-order PI^λ control system exhibits better control performance in terms of speed, stability, and system robustness.

4. Design of Fractional-Order PI^λ Control System Based on SSA

To reduce the unreliability of manual adjustment of PI^λ parameters and the difficulty of the fractional-order PI^λ controller design, the sparrow search algorithm is used in this section to perform parameter search for the fractional-order PI^λ controller. Finally, SSA is applied to the optimal design of the fractional-order Buck-Boost PI^λ control system to improve the system's performance.

4.1. Sparrow Search Algorithm

The sparrow search algorithm is a new swarm intelligence algorithm inspired by the interconversion between finder and joiner behaviors of sparrows during foraging. The discoverer in a sparrow colony is not only responsible for finding food for the whole population but also provides the direction and area of foraging. At the same time, the joiner relies on the discoverer to find food. The roles of finder and joiner can be flexibly changed. There are also aggressors in the population that monitor other efficient predators and attack efficient predators to increase their prey intake. In addition, birds outside the predation circle are vulnerable to natural predators; thus, they must constantly change their position to avoid them [26,27]. Sparrows in the center of the population will stay close to their peers and try to move to a safe area. When a predator is detected around the population, the population has an early warning signal and leaves the danger zone to feed in a safe area [28].

A population of n sparrows can be represented by the following equation.

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ \dots & \dots & \dots & \dots \\ x_{n,1} & x_{n,2} & \dots & x_{n,d} \end{bmatrix} \quad (21)$$

where n is the number of sparrows in a population and d is the dimensionality of the problem variable to be optimized. The fitness value of sparrows can be expressed as the following equation.

$$F(X) = \begin{bmatrix} f([x_{1,1} & x_{1,2} & \dots & x_{1,d}]) \\ f([x_{2,1} & x_{2,2} & \dots & x_{2,d}]) \\ \dots \\ f([x_{n,1} & x_{n,2} & \dots & x_{n,d}]) \end{bmatrix} \quad (22)$$

Discoverers with good fitness values prioritize access to food during foraging, and discoverers have a more extensive foraging search range than joiners because discoverers search for food for the population and provide foraging directions and areas for joiners. In each iteration, the position of the discoverer changes, as shown below.

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^t \cdot \exp(-\frac{i}{\gamma \cdot iter_{max}}) & \text{if } R_2 < ST \\ X_{i,j}^t + Q \cdot L & \text{if } R_2 \geq ST \end{cases} \quad (23)$$

where t is the number of current iterations and $j = 1, 2, \dots, d$, $iter_{max}$ is the constant for the maximum number of iterations. $X_{i,j}$ denotes the j th dimensional location information of the i th sparrow. γ is a random number, R_2 and ST are the warning and safety values, respectively, and Q denotes a random number obeying normal distribution. If $R_2 < ST$, it means that no predator is nearby, and the discoverer is able to conduct an extensive search. If $R_2 \geq ST$, it means that the early warning sparrow has found predators nearby and sends signals to other members of the population, all of which move quickly to forage elsewhere at this time.

The location of the joiners is updated as shown below.

$$X_{i,j}^{t+1} = \begin{cases} Q \cdot \exp(\frac{X_{worst} - X_{i,j}^t}{i^2}) & \text{if } i > n/2 \\ X_P^{t+1} + |X_{i,j}^t - X_P^t| \cdot A^+ \cdot L & \text{otherwise} \end{cases} \quad (24)$$

where X_{worst} is the current worst position within the global, X_P is the best position currently occupied by the discoverer, and A is a $1 \times d$ matrix. The elements in A are all 1 or -1 , $A^+ = A^T(AA^T)^{-1}$. If $i < n/2$, it means that the less adapted joiner did not acquire food because of the poor location and needed to forage elsewhere.

Assuming that sparrows sensing danger makes up 10% to 20% of the population and that the location of these sparrows is random, the mathematical expression is shown below.

$$X_{i,j}^{t+1} = \begin{cases} X_{best}^t + \mu \cdot |X_{i,j}^t - X_{best}^t| & \text{if } f_i < f_g \\ X_{i,j}^t + K \left(\frac{|X_{i,j}^t - X_{worst}^t|}{(f_i - f_w) + \epsilon} \right) & \text{if } f_i = f_g \end{cases} \quad (25)$$

where X_{best} is the current global optimal position, μ is the step control parameter. K is a random number taking values from -1 to 1 . f_g and f_w denote the current global best and worst fitness values, respectively, and ϵ is the smallest constant value that does not make the denominator zero.

In this algorithm, the objective function is set as follows:

$$F = \int_0^\infty (\omega_1 |e(t)| + \omega_2 u^2(t)) dt \quad (26)$$

where $e(t)$ is the error between the input value and the output value, taking into account the dynamic nature of the iterative process, taking the integration of its absolute value; $u(t)$ is the control value, which is added to avoid excessive control; ω_1 and ω_2 are the weights, and the value range is $[0, 1]$. In addition, restrictive measures are required to prevent

overshoot, that is, when an overshoot occurs, an additional overshoot term is introduced into the objective function, and the settings in this case are as follows:

$$F = \int_0^{\infty} (\omega_1 |e(t)| + \omega_2 u^2(t) + \omega_3 |e(t)|) dt, e(t) < 0 \quad (27)$$

where ω_3 is the weight and $\omega_3 \gg \omega_1$. In general, $\omega_1 = 0.999$, $\omega_2 = 0.001$, and $\omega_3 = 100$.

The SSA flow chart is shown in Figure 11.

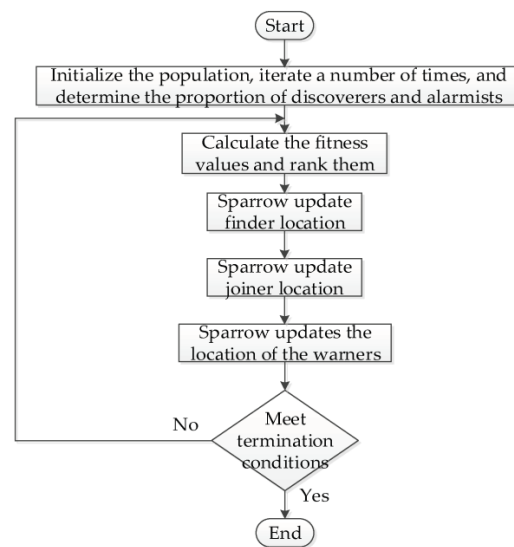


Figure 11. Sparrow search algorithm implementation process.

4.2. Design of Fractional-Order PI^λ Control System Based on SSA

In this section, the SSA is used to optimize the parameters of the fractional-order PI^λ controller in the fractional-order CCM converter circuit model to link the SSA with the fractional-order converter circuit model. The parameter values obtained from the sparrow and converter circuit simulation model can be used as the intermediate link between the SSA and the fractional-order converter circuit simulation model. The process of SSA to optimize the parameters of the fractional-order PI^λ controller is shown in Figure 12.

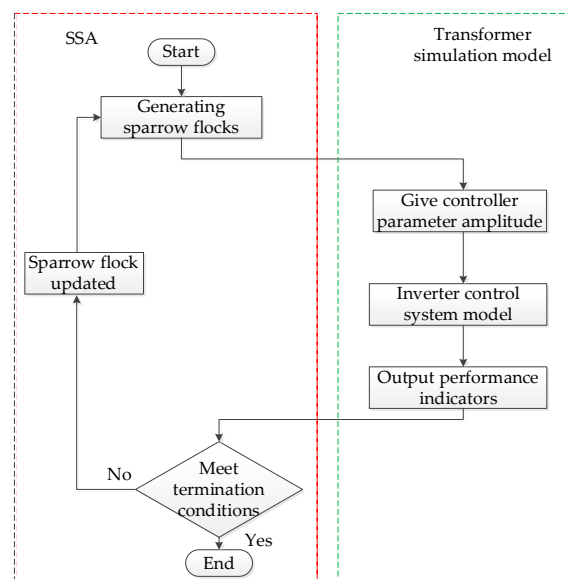


Figure 12. Sparrow search algorithm for parameter optimization of fractional order PI^λ controller.

Set the parameters of SSA: the number of sparrow flocks is $n = 40$, the dimension $d = 6$ represents each parameter of the PI^λ controller, and the maximum number of iterations is 100. The range of K_{PI} is 0.01–0.15, K_{II} is 5–15, K_{PV} is 0.02–0.18, K_{IV} is 10–30, λ_i is 0.8–1.1, and λ_v is 0.8–1.2. The controller parameters optimized with the algorithm are shown in Table 5.

Table 5. Optimized control system parameters.

K_{IP}	K_{PV}	K_{II}	K_{IV}	λ_i	λ_v
0.063	0.081	10.12	19.54	0.88	0.89

The fractional-order control system before controller parameter optimization and the Byrd plot after optimization using the SSA are shown in Figure 13.

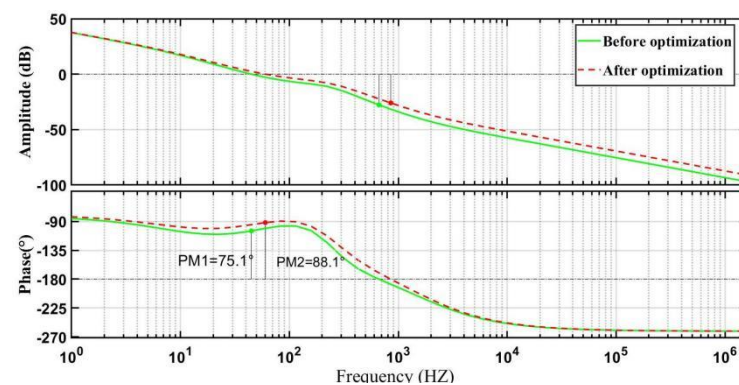


Figure 13. System Bode diagram of PI^λ controller optimized by SSA.

In Figure 13, the red dashed line shows the logarithmic curve of the system frequency characteristics under the optimized fractional-order PI^λ controller. The green solid line shows the logarithmic curve of the system frequency characteristics under the fractional-order PI^λ controller. From the figure, it can be seen that the phase margin with the fractional-order PI^λ controller is 75.1° and its shear frequency is 42.7 Hz; with the optimized fractional-order PI^λ control, the phase margin of the system is increased to 88.1° , and the shear frequency is 59.33 Hz. The optimized system has an improved phase margin and a corresponding increase in shear frequency. The optimized system has better speed, stability and robustness.

4.3. Simulation Analysis of the Optimized Control System

The simulation parameters of the system are shown in Table 5, and the simulation results are shown in Figure 14 when the resistive load is changed at a simulation time of 0.25 s.

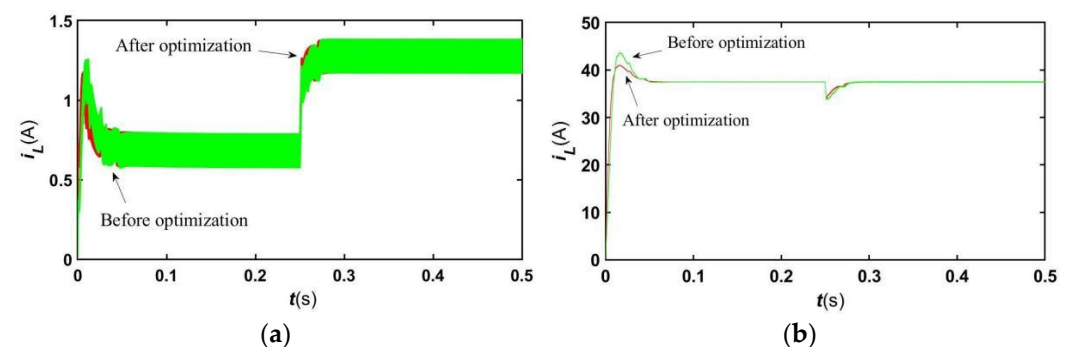


Figure 14. Comparison of inductor current and capacitor voltage before and after optimization of controller parameters. (a) Inductor current. (b) Capacitor voltage.

From Figure 14, it can be seen that the system after the optimization of the controller parameters has better performance in rise time, peak time, and anti-interference than the PI^λ control system before optimization, and the local simulation comparison enlargement is shown in Figures 15 and 16.

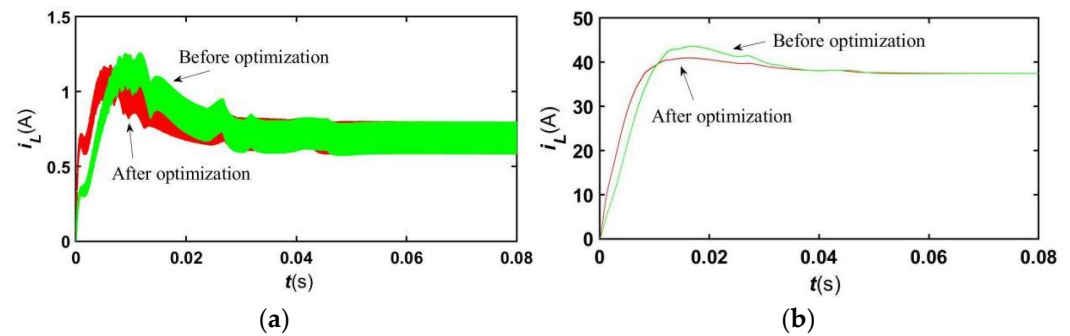


Figure 15. Enlarged view of inductor current and capacitor voltage in initial rising stage before and after optimization of controller parameters. (a) Inductor current. (b) Capacitor voltage.

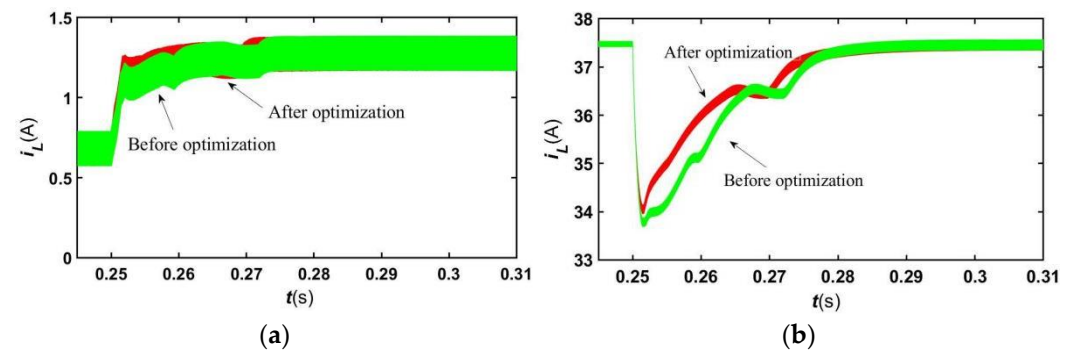


Figure 16. Enlargements of inductor current and capacitor voltage before and after controller parameter optimization when load changes suddenly. (a) Inductor current. (b) Capacitor voltage.

According to the simulation results in Figures 14–16, observing the initial phase and the process of sudden load change, the comparison of each performance index of the system before and after the optimization of the controller parameters is shown in the following table.

As shown in Tables 6 and 7, the system with optimized controller parameters reduces the rise time, regulation time, and overshoot amount. The regulation time of the initial state of the capacitor voltage is reduced by 11.58%, and the maximum fluctuation is reduced by 42.25%. After the disturbance, the regulation time of the initial state is reduced by 10.34%, the maximum fluctuation is reduced by 7.37%, and the system can reach the new steady state quickly. It is proved that the system performs better after controller parameter optimization than before optimization.

Table 6. Comparison of system response index of inductive current under two controllers.

Status	Controller	Rise Time (s)	Reduction Ratio of Adjustment Time (%)	Maximum Fluctuation Amount (A)	Maximum Volatility Reduction Ratio (%)
Initial state	PI	0.0041	—	0.461	—
	PI^λ	0.0025	39.34%	0.381	17.35%
Sudden load change	PI	0.0234	—	0	—
	PI^λ	0.0219	6.4%	0	0

Table 7. Comparison of system response index of capacitor voltage under two controllers.

Status	Controller	Rise Time (s)	Reduction Ratio of Adjustment Time (%)	Maximum Fluctuation Amount (A)	Maximum Volatility Reduction Ratio (%)
Initial state	PI	0.0095	—	6.13	—
	PI ^λ	0.0084	11.58%	3.54	42.25%
Sudden load change	PI	0.0232	—	3.8	—
	PI ^λ	0.0208	10.34%	3.52	7.37%

5. Conclusions

This paper establishes the fractional-order mathematical model and circuit model of the Buck-Boost converter under CCM based on fractional-order calculus theory. Based on the fact that inductors and capacitors are fractional-order by nature, the fractional-order model of the converter can be established to more accurately describe the real characteristics of the actual system and provide a more accurate theoretical basis for the optimal control of the converter. Secondly, the design of the PI^λ control system for a fractional-order converter is introduced, which adopts the control strategies of the current inner loop and voltage outer loop. The stability of the integer-order PI control system and the fractional-order PI^λ control system are compared. The fractional-order PI^λ controller is proved to have better performance than the integer-order controller by simulation. Finally, the parameters of the control system are optimized by the SSA algorithm, and the dynamic response performance and anti-interference performance of the fractional-order PI^λ control system before and after the optimization of the controller parameters are compared during the initial rise phase of the current and voltage and during the sudden load change. It is demonstrated by simulation that the system has better performance after controller parameter optimization than before optimization.

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